

AC2204 Marking Scheme

Introduction to Management Accounting · Winter 2021 · Paper 1

Examination Session and Year	Winter 2021
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Ivo De Loo
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings.

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Crafts Ltd	Marginal vs absorption costing, stock valuation reconciliation, arguments for each method	40
Q2 (Section B option)	Electro Ltd	Traditional (absorption) costing vs activity-based costing, evaluating ABC's value	30
Q3 (Section B option)	Angel Puppy Parlour	Scattergraph method, high-low method, CVP analysis, planning vs control	30
Q4 (Section B option)	Statement evaluation	Differential costs, step vs mixed costs, fixed cost per unit, conversion costs, cost classification	30

QUESTION 1**Crafts Ltd**

Total: 40 marks

Crafts Ltd commenced operations on 1 January 2020 and manufactures and sells a single product. Candidates are required to (A) prepare income statements under both marginal and absorption costing, (B) explain to the Managing Director why two different stock values and profit figures arise and which is the 'right' figure, and (C) discuss two arguments in support of each costing method.

Part	Requirement	Marks
(A)	Unit product cost, closing stock value and operating profit – marginal and absorption costing	18
(B)	Explanatory note addressing which stock value is 'right', with profit reconciliation	10
(C)	Two arguments in support of each costing method	12

Question 1(A)**18 marks**

Crafts Ltd commenced operations 1 Jan 2020. Sales 2,000 units; Selling price €35.00; Production 2,800 units; Variable manufacturing expenses (total) €61,600; Fixed manufacturing expenses (total) €14,280; Variable selling expenses (total) €10,000; Fixed selling expenses (total) €5,000. (A) Calculate the unit product cost, the value of closing stock and the operating profit using (i) Marginal costing and (ii) Absorption costing.

Full Model Answer**Preliminary workings**

Closing stock = Production – Sales = 2,800 – 2,000 = 800 units. Variable manufacturing cost per unit = €61,600 ÷ 2,800 = €22.00. Fixed manufacturing overhead per unit (absorption costing only) = €14,280 ÷ 2,800 = €5.10.

(i) Marginal costing**Unit product cost and closing stock - marginal costing**

Item	€
Unit product cost (variable manufacturing cost only)	22.00
Closing stock value (800 units × €22.00)	17,600

Income statement - marginal costing

Item	€
Sales revenue (2,000 units × €35.00)	70,000
Less: Variable cost of goods sold (2,000 units × €22.00)	(44,000)
Less: Variable selling expenses	(10,000)
Contribution margin	16,000
Less: Fixed manufacturing expenses	(14,280)
Less: Fixed selling expenses	(5,000)
Operating loss (marginal costing)	(3,280)

(ii) Absorption costing**Unit product cost and closing stock - absorption costing**

Item	€
Unit product cost (€22.00 variable + €5.10 fixed)	27.10
Closing stock value (800 units × €27.10)	21,680

Income statement - absorption costing

Item	€
Sales revenue (2,000 units × €35.00)	70,000
Less: Cost of goods sold (2,000 units × €27.10)	(54,200)
Gross profit	15,800
Less: Variable selling expenses	(10,000)
Less: Fixed selling expenses	(5,000)
Operating profit (absorption costing)	800

Notable result: Marginal costing shows an operating **loss** of €3,280 for the year, while absorption costing shows an operating **profit** of €800 — the two methods do not merely differ in amount here, they disagree on whether the company made a profit or a loss at all. This is explored further in part (B).

Mark Allocation

Marks	Criteria
16-18	Both unit product costs, both closing stock values, and both fully-formatted income statements correct, with contribution margin and gross profit subtotals clearly shown, including the correct sign of the marginal costing loss.
12-15	Substantially correct with a minor computational error in one figure, or one method fully correct and the other largely correct.
7-11	Correct approach to at least one costing method; some errors or omissions in workings or presentation.
0-6	Limited understanding of the distinction between the two costing methods; minimal correct workings.

Question 1(B)**10 marks**

The Managing Director does not understand how there can be two different stock values reported for the same period and wishes to know which is the 'right' figure. Prepare a short note, including a reconciliation of the profit figures, which addresses this query.

Full Model Answer**Why two different stock values arise**

Marginal costing values closing stock at variable manufacturing cost only (€22.00 per unit), treating fixed manufacturing overhead as a period cost charged in full against the year it is incurred. Absorption costing values closing stock at full production cost, including a share of fixed manufacturing overhead (€27.10 per unit), on the basis that fixed overhead is a necessary cost of production and should be matched against the revenue earned when the unit is eventually sold. Neither figure is 'wrong' as an arithmetical matter — they are simply the result of two different, internally consistent conventions for treating fixed manufacturing overhead.

Which is the 'right' figure?

There is no single universally 'right' answer — the appropriate figure depends on the purpose. For external financial reporting, IAS 2 *Inventories* requires absorption costing, so €27.10 per unit (€21,680 total) is the figure that must appear in Crafts Ltd's statutory financial statements. For internal decision-making and understanding the underlying trading performance of the business this year, marginal costing is often considered more informative: it is not affected by the level of production relative to sales, and it reveals that, on a purely trading basis, more was spent on the year's costs than was recovered through sales revenue (an underlying loss of €3,280), a fact that is masked in the absorption costing profit of €800 because a large volume of unsold production (800 units) is carrying €4,080 of fixed overhead forward into next year's stock.

Profit reconciliation

Item	€
Operating loss – marginal costing	(3,280)
Add: Fixed manufacturing overhead carried forward in closing stock (800 units × €5.10)	4,080
Less: Fixed manufacturing overhead carried forward in opening stock (none – first year of trading)	0
Operating profit – absorption costing	800

Caution for the Managing Director: The absorption costing profit of €800 should not be read as evidence that the business is performing well: it depends entirely on the 800 units of unsold production carrying €4,080 of cost forward. If those units are never sold, or are sold at a loss next year, the apparent profit shown this year will not be realised. This is a common criticism of absorption costing: it can create an incentive to over-produce simply to defer fixed costs into stock and inflate reported profit.

Mark Allocation

Marks	Criteria
9–10	Clear explanation of why the two stock values differ, a balanced and well-reasoned answer to 'which is right' (recognising both the IAS 2 requirement and the internal decision-making perspective), and a correct profit reconciliation.
6–8	Good explanation and mostly correct reconciliation; the 'which is right' discussion is one-sided or less developed.
3–5	Some understanding shown of why the figures differ, but explanation is generic or the reconciliation is materially incomplete.
0–2	No meaningful explanation or reconciliation provided.

Question 1(C)**12 marks**

Management at Crafts Ltd is undecided about choosing between marginal costing and absorption costing. Discuss two arguments in support of the use of each of the methods.

Full Model Answer**Arguments in support of marginal costing**

1. Reveals the true underlying trading performance: as shown in part (B), marginal costing profit is unaffected by changes in stock levels and depends only on units sold, giving management a clearer, less distortable picture of how the business actually performed during the year — important where, as here, production significantly exceeded sales. **2.**

Directly supports short-term decision-making: by separating variable from fixed costs and reporting contribution margin, marginal costing feeds directly into CVP analysis, breakeven calculations, and relevant costing decisions (special orders, make-or-buy) without the distortion of arbitrarily allocated fixed overhead.

Arguments in support of absorption costing

1. Required for external financial reporting: IAS 2 *Inventories* requires inventory to be valued at full production cost (including a fair allocation of fixed overhead) in the statutory financial statements, so Crafts Ltd must use absorption costing for external reporting purposes regardless of what is used internally. **2. Recognises that fixed manufacturing overhead is a genuine cost of production:** the factory, machinery and supervision needed to manufacture 2,800 units is a real cost of producing those units, and absorption costing ensures this cost is reflected in the cost of each unit, supporting long-term pricing decisions that need to recover the full cost of production, not just variable cost.

Mark Allocation

Marks	Criteria
10-12	Two distinct, well-explained arguments given for each method (four arguments in total), correctly characterising the practical strengths of marginal and absorption costing respectively, with reference to the specific figures in this question where relevant.
7-9	Good arguments given for both methods; one argument less developed or slightly generic.
4-6	At least one reasonable argument given for each method, or strong arguments for only one method.
0-3	Minimal or no relevant arguments given.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2**Electro Ltd**

Total: 30 marks

Management at Electro Ltd are concerned about the product cost information generated by the conventional (absorption) costing system currently in use. Candidates are required to (A) recalculate product costs under both traditional and activity-based costing, and (B) evaluate a sceptical statement about whether ABC is worth the effort.

Part	Requirement	Marks
(A)	Total costs and cost per unit – traditional costing and activity-based costing	22
(B)	Agree/disagree discussion: is ABC worth the time it takes to complete?	8

Question 2(A)**22 marks**

Electro Ltd absorbs production overheads (€152,400) on the basis of machine hours, at €48 per machine hour. Direct labour costs €26 per hour. Product A: 1,000 units, €15 DM/unit, 1 MH/unit, 1 DLH/unit. Product B: 900 units, €18 DM/unit, 2.25 MH/unit, 0.5 DLH/unit. Product C: 300 units, €26 DM/unit, 0.5 MH/unit, 3 DLH/unit. ABC cost pools: Maintenance 30% (machine hours), Production scheduling 25% (set-ups), Quality 30% (units), Materials handling 15% (purchase orders). Purchase orders: A=5, B=5, C=10 (total 20). Set-ups: A=5, B=9, C=6 (total 20). (A) Calculate the total costs and the cost per unit for each product, using (i) Conventional (absorption) costing and (ii) Activity-based costing.

Full Model Answer - (i) Conventional (absorption) costing

Overhead absorption rate = €152,400 ÷ 3,175 total machine hours (1,000 × 1 + 900 × 2.25 + 300 × 0.5 = 1,000 + 2,025 + 150) = €48 per machine hour, as given.

Conventional (absorption) costing - cost per unit

Item	Product A	Product B	Product C
Direct materials	15.00	18.00	26.00
Direct labour (DLH × €26)	26.00	13.00	78.00
Overhead (MH/unit × €48)	48.00	108.00	24.00
Total cost per unit	89.00	139.00	128.00

Conventional (absorption) costing - total costs

Item	Product A (€)	Product B (€)	Product C (€)
Units produced	1,000	900	300
Total cost (unit cost × units)	89,000	125,100	38,400

Check: Total overhead absorbed = €48,000 (A) + €97,200 (B) + €7,200 (C) = €152,400, agreeing with the total production overhead given.

Full Model Answer - (ii) Activity-based costing

Working - cost pools and activity rates

Cost pool	Pool amount (€)	Cost driver	Total activity	Rate
Maintenance (30%)	45,720	Machine hours	3,175 MH	€14.40 per MH
Production scheduling (25%)	38,100	No. of set-ups	20 set-ups	€1,905.00 per set-up
Quality (30%)	45,720	No. of units	2,200 units	€20.78 per unit
Materials handling (15%)	22,860	No. of purchase orders	20 orders	€1,143.00 per order
Total	152,400			

Working - overhead allocated to each product

Cost pool	Product A (€)	Product B (€)	Product C (€)
Maintenance (total MH × €14.40)	14,400	29,160	2,160
Scheduling (set-ups × €1,905)	9,525	17,145	11,430
Quality (units × €20.78)	20,782	18,704	6,235
Materials handling (orders × €1,143)	5,715	5,715	11,430
Total overhead	50,422	70,724	31,255

Check: Total overhead allocated = €50,422 + €70,724 + €31,255 = €152,400 (rounding), agreeing with the total production overhead given.

Activity-based costing - total costs and cost per unit

Item	Product A	Product B	Product C
Direct materials (total)	15,000	16,200	7,800
Direct labour (total)	26,000	11,700	23,400
Overhead (from above)	50,422	70,724	31,255
Total cost	91,422	98,624	62,455
Units produced	1,000	900	300
Cost per unit	91.42	109.58	208.18

Comparison: ABC increases the unit cost of Product A slightly (€89.00 → €91.42) and Product B slightly (€139.00 → €109.58 is actually a decrease) while dramatically increasing Product C's unit cost (€128.00 → €208.18, a 63% increase). Product C, despite its low machine-hour usage per unit, consumes a disproportionate share of materials handling activity (10 of the 20 total purchase orders for only 300 units) which the traditional machine-hour-based method fails to capture, resulting in Product C being significantly undercosted under conventional costing.

Mark Allocation

Marks	Criteria
18-22	Traditional costing correctly calculated; all four ABC cost pool rates correctly calculated; overhead correctly allocated to each product; ABC total cost and cost per unit correct for all three products.
13-17	Substantially correct with minor computational errors in one or two pool rates or allocations.
7-12	Traditional costing correct; ABC approach understood but rates or allocations contain multiple errors.
0-6	Limited understanding of either costing method; minimal correct workings.

Question 2(B)**8 marks**

Consider the following statement: "Activity-based costing is just a more complicated way of dividing costs between products. Total costs don't change anyway, so ABC isn't worth the time it takes to complete." State whether you agree or disagree, discussing reasons for your answer.

Full Model Answer**DISAGREE**

It is true that total overhead (€152,400) is identical under both methods — ABC does not create or destroy cost, only reallocates it. However, the conclusion that ABC therefore 'isn't worth the time' does not follow from this observation.

Why the total staying the same does not make the allocation irrelevant

Although the company-wide total is unchanged, the *distribution* of that total across individual products changes materially, as shown in part (A): Product C's cost per unit rises from €128.00 to €208.18 (a 63% increase), revealing that traditional costing was significantly undercosting Product C and, correspondingly, overcosting Product A and B relative to their actual resource consumption. Decisions such as pricing, product mix, and whether to continue or discontinue a product are made at the *individual product* level, not at the company-wide total level — so a more accurate individual-product cost has real decision-making value even though the total is unchanged.

The genuine risk of continuing to use the distorted traditional figures

If Electro Ltd continues pricing or evaluating Product C based on its undercosted traditional figure of €128.00, it risks underpricing the product, over-promoting it relative to its true profitability, or failing to recognise that Product A and B are being over-costed and potentially over-priced relative to competitors. The cost of the ABC exercise should be weighed against the value of avoiding these kinds of decision errors, not dismissed purely because the overall total is unchanged.

A fair caveat to the disagreement

It is reasonable to note that ABC does involve a genuine cost-benefit trade-off: for a business with simple, homogeneous products or low overhead relative to total cost, the effort of maintaining an ABC system may not be justified. For a company like Electro Ltd, however, with three products of very different complexity and resource consumption patterns, the improvement in costing accuracy revealed above suggests ABC is likely to be worthwhile despite the extra effort involved.

Mark Allocation

Marks	Criteria
7-8	Clear disagreement with the statement's conclusion; correctly distinguishes between the unchanged total and the materially changed product-level allocation; references the specific cross-subsidization revealed in part (A); acknowledges the genuine cost-benefit trade-off involved in adopting ABC.
5-6	Correct disagreement with good reasoning; limited or no reference to the specific figures from part (A).
2-4	Some relevant discussion but generic, or an unbalanced one-sided argument without acknowledging any merit in the original statement.
0-1	Agrees with the statement without meaningful justification, or no meaningful discussion provided.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3**Angel Puppy Parlour**

Total: 30 marks

Angela French owns a mobile pet grooming service and needs projected costs for a meeting with her credit union. Candidates are required to (A) apply the scattergraph method, (B) apply the high-low method and full CVP analysis for a December forecast, with an explanation of the key terms used, and (C) distinguish planning from control activities in the context of Angela's business.

Part	Requirement	Marks
(A)	Scattergraph method – approximate fixed cost and variable cost per client	8
(B)	High-low cost function, breakeven point, profit/loss and margin of safety for December, with explanation of each term's use	16
(C)	Distinguish planning and control activities, applied to Angela's business	6

Question 3(A)**8 marks**

Monthly activity level (clients) and total costs: June 80, €4,100; July 130, €5,100; August 60, €3,700; September 30, €3,100; October 40, €3,300; November 100, €4,500. (A) Prepare a scatter graph using the data provided (graph paper not required). Using your graph, determine the approximate monthly fixed cost and approximate variable cost per client.

Full Model Answer**Method**

Each month's data (number of clients on the horizontal axis, total cost on the vertical axis) is plotted as a point on a graph. A straight line is then drawn through the data by eye, positioned to best represent the overall trend of the six points (rather than necessarily passing through any two specific points, as in the high-low method). The point where this line crosses the vertical axis (at zero clients) gives the approximate monthly fixed cost; the slope of the line (the additional cost for each additional client) gives the approximate variable cost per client.

Data points to be plotted

Month	Clients (x-axis)	Total cost, € (y-axis)
September	30	3,100
October	40	3,300
August	60	3,700
June	80	4,100
November	100	4,500
July	130	5,100

Approximate result: This data set happens to lie in an almost perfectly straight line, so a well-drawn line of best fit gives an approximate y-intercept (fixed cost) of around €2,500 per month and a slope (variable cost per client) of around €20. Because this is a visual, hand-drawn estimation method, candidates' figures are marked with reasonable tolerance (approximately €2,200–€2,800 fixed cost, €18–€22 variable cost per client) rather than requiring an exact figure — the precise figures are confirmed algebraically using the high-low method in part (B).

Mark Allocation

Marks	Criteria
7-8	Clear description of the scattergraph method and how the intercept and slope are read from the graph; a reasonable fixed cost and variable cost per client estimate (within the tolerance range) with correct reasoning.
5-6	Good description of the method with a reasonable estimate; minor imprecision in the figures.
2-4	General understanding of the scattergraph approach shown, but the description or estimate is vague or significantly off the expected range.
0-1	No meaningful attempt at the scattergraph method.

Question 3(B)**16 marks**

Angela anticipates serving 150 clients in December at a sales price of €40 per client. (i) Using the high-low method estimate a cost function for the business and determine the estimated total costs for December. (ii) Determine the budgeted breakeven point in units and sales revenues for December. (iii) Determine the estimated profit or loss and the percentage margin of safety for December. (iv) Explain the use of each of the underlined terms above to Angela in the management of the business.

Full Model Answer – (i) High-low cost function**Working – identify high and low points**

Month	Clients	Total cost (€)
July (highest activity)	130	5,100
September (lowest activity)	30	3,100
Difference	100	2,000

Cost function

Item	Value
Variable cost per client ($€2,000 \div 100$)	€20.00 per client
Fixed cost (at high point: $€5,100 - (130 \times €20)$)	€2,500
Fixed cost (at low point: $€3,100 - (30 \times €20)$) – check	€2,500
Cost function	Total cost = €2,500 + (€20 × clients)

Estimated total costs for December (150 clients)

Item	Value
$€2,500 + (€20 \times 150)$	€2,500 + €3,000
Estimated total costs, December	€5,500

Consistency with part (A): This data set is close to perfectly linear, so the high-low figures (€2,500 fixed, €20 variable per client) closely match the scattergraph estimate from part (A), which is a useful consistency check.

Full Model Answer - (ii) Budgeted breakeven point**Working**

Item	Value
Selling price per client	€40.00
Variable cost per client (from high-low)	€20.00
Contribution margin per client	€20.00
Fixed cost (December, from cost function)	€2,500

Breakeven point

Item	Value
Breakeven units ($€2,500 \div €20$)	125 clients
Breakeven revenue ($125 \times €40$, or $€2,500 \div 50\%$ CM ratio)	€5,000

Full Model Answer - (iii) Estimated profit and margin of safety**Working - December profit**

Item	€
Sales revenue ($150 \text{ clients} \times €40$)	6,000
Less: Estimated total costs (from part (i))	(5,500)
Estimated operating profit	500

Cross-check via CVP: $150 \text{ clients} \times €20 \text{ contribution per client} = €3,000 \text{ total contribution}$; $€3,000 - €2,500 \text{ fixed costs} = €500 \text{ operating profit}$, confirming the figure above.

Margin of safety

Item	Value
Margin of safety % ($(150 - 125) \div 150$)	16.67%

Full Model Answer - (iv) Explaining the terms to Angela

Cost function

The cost function (Total cost = €2,500 + €20 × clients) allows Angela to predict her total costs at any level of activity, not just the specific months already observed. This is directly useful for the credit union meeting: she can provide a projected cost figure for December (or any other future month) with a clear, defensible basis, rather than a rough guess.

Breakeven point (units and sales revenue)

The breakeven point (125 clients, or €5,000 in sales revenue) tells Angela the minimum level of activity she must achieve in December before the business starts to make a profit. This is useful for setting a minimum monthly client target and for assessing how achievable her forecast of 150 clients is relative to that minimum — here, the forecast comfortably exceeds breakeven, which is a positive signal to present to the credit union.

Margin of safety

The margin of safety (16.67%) tells Angela how far actual client numbers could fall short of the 150-client forecast before the business would start to make a loss. This gives her (and the credit union) a sense of how much 'cushion' there is in the forecast — a relatively modest margin of safety here suggests the December forecast, while profitable, does not have a large buffer against a shortfall in demand.

Mark Allocation

Marks	Criteria
13-16	High-low cost function, breakeven point (units and revenue), and profit/margin of safety all correctly calculated; all key terms explained specifically in terms of their practical use to Angela (not generic definitions).
9-12	Substantially correct calculations with a minor error; explanations of the terms are reasonable but less specifically tied to Angela's situation.
5-8	Correct general approach with some errors in the calculations; explanations largely generic/textbook rather than applied.
0-4	Limited understanding of the high-low method or CVP calculations; minimal correct workings or explanation.

Question 3(C)**6 marks**

Distinguish between planning activities and control activities in management accounting, outlining how each can be of use to Angela in managing her business.

Full Model Answer**Planning**

Planning is the process of establishing objectives and determining the means by which those objectives will be achieved; it is forward-looking and includes activities such as budgeting and forecasting. *Application to Angela:* the December forecast prepared in part (B) — projecting 150 clients, €6,000 revenue, €5,500 costs, and €500 profit — is a planning activity. It allows Angela to set targets for the month ahead and to present a credible, well-supported case to the credit union about the business's expected performance.

Control

Control is the process of comparing actual results against the plan, identifying variances, and taking corrective action where necessary; it is a monitoring and feedback process that follows planning. *Application to Angela:* once December has passed, Angela should compare her actual number of clients and actual costs against the 150-client, €5,500-cost budget prepared in part (B). If actual clients fall significantly short of 125 (the breakeven point), or if actual costs run well above the €20 variable/€2,500 fixed cost function, this signals a problem requiring investigation and corrective action (e.g. increased marketing, cost review) before the situation worsens.

The planning-control cycle: Planning and control are not one-off, separate activities but form a continuous cycle: control information (actual results) feeds back into the next period's planning (revised forecasts and budgets), helping Angela's cost and revenue estimates become more accurate over time as she gathers more months of actual data.

Mark Allocation

Marks	Criteria
5-6	Both planning and control clearly and accurately distinguished, each with a specific, well-reasoned application to Angela's business referencing the figures from part (B).
3-4	Both concepts distinguished accurately with reasonable application to Angela's business.
1-2	One concept explained well; the other explained only partially or generically, or without application to Angela's business.
0	No meaningful distinction drawn.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4**Statement Evaluation**

Total: 30 marks

Five independent statements covering differential costs, step vs mixed costs, fixed cost per unit behaviour, conversion cost calculations, and the situational nature of cost classification. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Differential costs and cost behaviour	6
(B)	Step costs vs mixed costs	6
(C)	Fixed cost per unit as production increases	6
(D)	Galway Ltd – manufacturing overhead from conversion cost percentage	6
(E)	Situational nature of cost classification	6

Question 4(A)**6 marks**

Only variable costs can be differential costs.

Full Model Answer**DISAGREE**

A differential cost is the difference in total cost between two alternative courses of action. This difference can arise from either variable **or** fixed costs — the classification of a cost as fixed or variable is entirely independent of whether it is differential.

Why fixed costs can be differential too

A fixed cost is differential (relevant) whenever it differs between the alternatives being compared, regardless of its fixed cost-behaviour label. For example, if one alternative under consideration requires renting additional space or hiring an extra supervisor (as with the storage rental and production space decisions seen in earlier questions), and the other alternative does not, that fixed cost is directly differential between the two alternatives, even though it does not vary with output volume.

What actually determines whether a cost is differential

A cost is differential simply if it changes depending on which alternative is chosen; it is not differential (i.e. it is irrelevant to the decision) only if it is identical under all alternatives being considered — which can be true of some variable costs (e.g. a cost that both alternatives share identically) just as it can be true of some fixed costs.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with a clear explanation that differential cost status depends on whether a cost differs between alternatives, not on whether it is fixed or variable, supported by a relevant example.
3-4	Correct disagreement with reasonable explanation but no clear illustrative example.
1-2	Correct verdict stated but with minimal or unclear explanation.
0	Incorrect verdict or no meaningful explanation.

Question 4(B)**6 marks**

Step costs and mixed costs are just different names for the same thing.

Full Model Answer**DISAGREE**

Step costs and mixed costs describe two different patterns of cost behaviour and are not interchangeable terms.

Step costs

A step cost (or step-fixed cost) remains constant within a narrow range of activity, then jumps to a new, higher level once a particular activity threshold is exceeded, before remaining flat again over the next range. Graphed against volume, a step cost looks like a series of horizontal steps. *Example:* supervisor salaries, where one supervisor can oversee up to a certain number of staff or clients before an additional supervisor must be hired.

Mixed (semi-variable) costs

A mixed cost contains both a fixed element and a variable element that increases continuously and smoothly with activity — it does not jump in discrete steps. Graphed against volume, a mixed cost is a single straight line with a positive vertical intercept (the fixed element) and a constant positive slope (the variable element). This is exactly the cost behaviour identified for Angela's grooming business in Question 3, where costs rise smoothly and continuously with the number of clients, not in discrete jumps.

Mark Allocation

Marks	Criteria
5-6	Correct disagreement with accurate, clearly distinguished descriptions of both step costs and mixed costs, including how each behaves graphically, with a relevant example for at least one.
3-4	Correct disagreement with reasonable descriptions of both terms; example(s) missing or the graphical distinction not explained.
1-2	Correct disagreement but with a vague or partially inaccurate explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(C)**6 marks**

As production increases within the relevant range, fixed costs per unit decrease.

Full Model Answer**AGREE**

Total fixed costs remain constant within the relevant range of activity, regardless of the level of production. Because fixed cost per unit is calculated as total fixed cost divided by the number of units produced, as production increases, the same total fixed cost is spread over more units, so the fixed cost per unit falls.

Illustration using this paper's own figures

In Question 1, Crafts Ltd's total fixed manufacturing overhead of €14,280 gives a fixed cost per unit of €5.10 when spread over 2,800 units produced. If production had instead been 3,600 units (with the same €14,280 total fixed cost), the fixed cost per unit would fall to €3.97 ($€14,280 \div 3,600$) — the total fixed cost is unchanged, but the per-unit figure falls as volume rises. This relationship only holds within the relevant range; outside this range, fixed costs may 'step up' to a new, higher total (as illustrated by step costs in part (B)).

Mark Allocation

Marks	Criteria
5-6	Correct agreement with clear explanation of why fixed cost per unit falls as volume rises, including a specific illustrative example (ideally drawn from elsewhere in the paper) and reference to the relevant range.
3-4	Correct agreement with reasonable explanation; illustrative example or relevant range qualification missing.
1-2	Correct agreement but with minimal or unclear explanation.
0	Incorrect verdict or no meaningful explanation.

Question 4(D)**6 marks**

Galway Ltd's manufacturing overhead is 55% of its total conversion costs. If direct labour is €40,500 and direct materials are €45,860, then the manufacturing overhead is €49,500.

Full Model Answer**AGREE**

The manufacturing overhead figure of €49,500 given in the statement is correct.

Key point - direct materials is not used in this calculation

Conversion cost is defined as direct labour plus manufacturing overhead (the cost of converting raw materials into a finished product); it deliberately **excludes** direct materials. The €45,860 direct materials figure given in the statement is not required for this calculation — it is additional information not relevant to finding manufacturing overhead from the conversion cost percentage.

Working

Item	Value
Let manufacturing overhead (MOH) = 55% of conversion cost	$MOH = 0.55 \times (DL + MOH)$
$MOH = 0.55 \times €40,500 + 0.55 \times MOH$	$MOH - 0.55 MOH = €22,275$
$0.45 \times MOH = €22,275$	$MOH = €22,275 \div 0.45$
Manufacturing overhead	€49,500

Check

Item	€
Direct labour	40,500
Manufacturing overhead	49,500
Conversion cost	90,000
Manufacturing overhead as % of conversion cost ($€49,500 \div €90,000$)	55% - confirms the given figure

Mark Allocation

Marks	Criteria
5-6	Correct agreement; conversion cost correctly defined as excluding direct materials; algebraic working correctly solved to give €49,500, with a check confirming the 55% relationship.
3-4	Correct agreement and correct final figure, but the exclusion of direct materials from conversion cost not explicitly explained.
1-2	Correct verdict stated but workings contain an error or direct materials incorrectly included in the calculation.
0	Incorrect verdict or no meaningful workings.

Question 4(E)**6 marks**

Classification of costs into fixed and variable depends on the decision situation.

Full Model Answer**AGREE**

Whether a cost behaves as fixed or variable is not an absolute, universal property of the cost itself — it depends on the specific decision, time horizon, and relevant range being considered.

Time horizon

In the short run, many costs behave as fixed (e.g. a supervisor's salary, equipment lease payments) because they cannot easily be adjusted regardless of the level of activity. Over a longer time horizon, however, virtually all costs become variable/adjustable as management can change capacity, renegotiate contracts, or restructure operations — the same cost can therefore be classified differently depending on the time frame relevant to the decision being made.

The specific decision and relevant range

A cost's classification can also depend on which activity or cost driver is being used to assess it, and the relevant range of activity under consideration. For example, machine maintenance cost might be treated as fixed for the purposes of a short-term pricing decision within the normal range of activity, but would be treated as a step cost or even variable if the decision under consideration involves a much larger change in production volume that would require additional machinery.

Mark Allocation

Marks	Criteria
5-6	Correct agreement with a well-developed explanation covering at least two distinct reasons why cost classification is situational (e.g. time horizon and relevant range/decision context), with a relevant example.
3-4	Correct agreement with one well-explained reason and a supporting example.
1-2	Correct agreement but with minimal or generic explanation.
0	Incorrect verdict or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS